18 October 2026

Dear Editors of PLOS ONE,

Re: **PONE-D-26-50597 — author-prepared updated manuscript**

I have prepared an update to my manuscript submitted on 9 October 2026. The proposed updated title is **"Source-cohort expansion improves weaker-class coverage in cross-subject motor-imagery EEG decoding."** Please advise whether you would like the completed additional experiment incorporated into the manuscript under consideration. The accompanying files are an author-prepared candidate pending your instructions.

The completed experiment compares 9, 29 and 60 source participants using common 18-channel, dimensionless processing and participant-held-out source validation, followed by evaluation on 85 participants from a separate acquisition programme. Each neural trajectory received the same maximum of 5,120 optimizer updates, making computational allocation explicit alongside source composition and available trials.

In the primary BROAD/tail-minimum-recall comparison, S60 increased mean weaker-class recall from 27.31% to 48.48% relative to S9: a 21.17-percentage-point difference (95% CI 17.12–25.21); 72 of 85 participants improved. Balanced accuracy increased by 5.85 points. Descriptive follow-up analyses show a 21.72-point weaker-class gain for S29 over S9 and a 13.82-point gain for conventional CSP from S9 to S60. All prespecified comparisons accompany these findings.

The contribution is a fixed-budget evaluation of source-cohort expansion and participant-level class coverage, with source-only checkpoint selection and a reproducible evidence chain. Frozen code, predictions and source-selection records are public; independent saved-prediction auditing passed 1,179 checks. The external analysis is identified as exploratory because historical first-use independence remains uncertain.

This secondary public-data study recruited no new participants. I confirm that Chongqing Medical University's research administration office confirmed on 17 October 2026 that the BNCI2014-001, Yang and Dreyer analysis was exempt from ethics review under Article 32(1) of the 2023 national measures. Ethics, data availability, contributions and AI assistance are reported. Third-party recordings remain with their providers; research materials are public on GitHub.

Thank you for considering this work.

Sincerely,

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